

Composing Functions from Symbolic Rules

Composing at a number, composing into a new rule, comparing the two orders, and solving an equation built from a composition

The model: two machines in a row

A composition sends the output of one machine straight into the next. For $f(x) = 2x + 5$ and $g(x) = x^2 - 3$, the chain $x \rightarrow \boxed{g} \rightarrow \boxed{f} \rightarrow \text{output}$ is what $(f \circ g)(x) = f(g(x))$ means. Follow one column at a time:

input x	after g : $x^2 - 3$	after f : $2(\cdot) + 5$
0	-3	-1
1	-2	1
3	6	17

Directions

- Read $(f \circ g)(x)$ as “ g first, then f ”: the function written closest to x runs first.
- To compose *symbolically*, substitute the whole rule for g wherever f ’s formula has an x , then simplify.
- Keep the substituted rule in brackets while you simplify — that is where most sign and squaring errors come from.

1. Let $f(x) = 2x + 5$ and $g(x) = x^2 - 3$, the two machines in the table above.

(a) Find $g(4)$: _____ (b) Find $f(g(4))$: _____

2. Let $f(x) = 3x - 1$ and $g(x) = x + 4$. Find the rule for $(f \circ g)(x)$, simplified.

Answer: _____

3. Still with $f(x) = 3x - 1$ and $g(x) = x + 4$.

(a) Find the rule for $(g \circ f)(x)$, simplified: _____

(b) Compare your answer with $(f \circ g)(x)$ from problem 2. Do the two orders give the same function? Explain what causes the difference.

4. Let $f(x) = x^2$ and $g(x) = x - 6$.

(a) Find $(f \circ g)(2)$: _____

(b) Find $(g \circ f)(2)$: _____

5. Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find the rule for $(f \circ g)(x)$, expanded and simplified.

Answer: _____

6. Let $f(x) = 2x + 5$ and $g(x) = x - 1$. Solve $(f \circ g)(x) = 17$ for x .

Answer: _____

7. Let $f(x) = \frac{x}{x+1}$ and $g(x) = \frac{2}{x}$. Find the rule for $(f \circ g)(x)$, simplified into a single fraction.

Answer: _____

8. Let $f(x) = x^3$ and $g(x) = x + 1$.

(a) Find the rule for $(f \circ g)(x)$, expanded: _____

(b) Find $(g \circ f)(2)$: _____

(c) Use part (a) at $x = 2$ together with part (b) to explain why the order of a composition changes the function, not just this one value.

9. Let $f(x) = x^2$ and $g(x) = x + 3$. Solve $(f \circ g)(x) = 49$ for x .

Answer: _____

10. Challenge. Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 5$.

(a) Find the rule for $(f \circ g)(x)$, simplified: _____

(b) Find $(f \circ g)(3)$, rounded to two decimal places: _____

(c) The function f by itself is undefined for many inputs, yet $(f \circ g)(x)$ works for every real number x . Explain why.